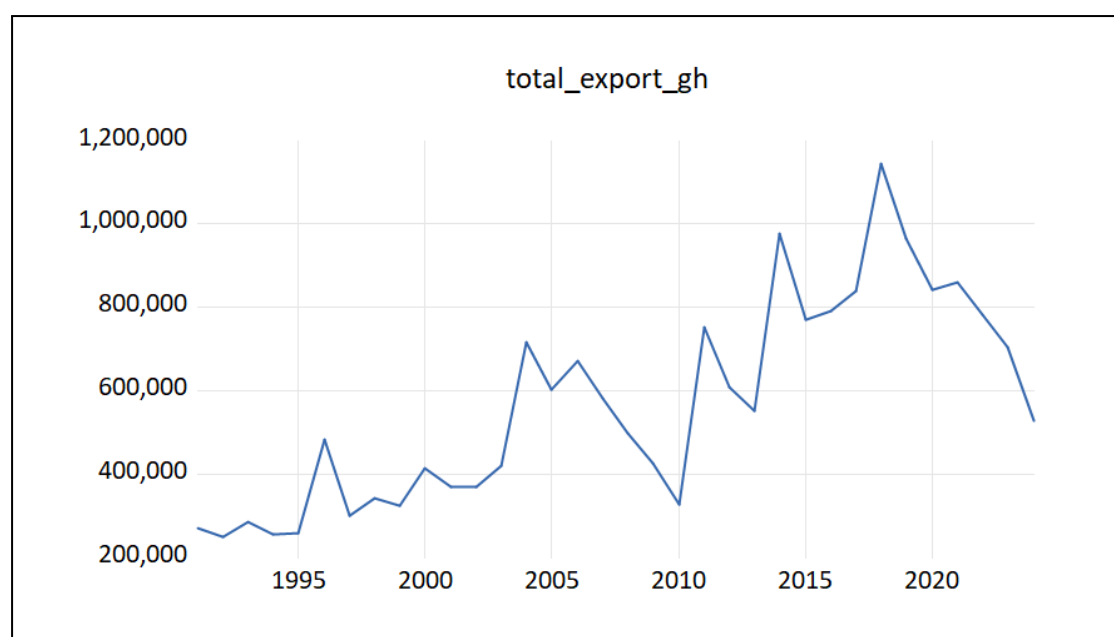
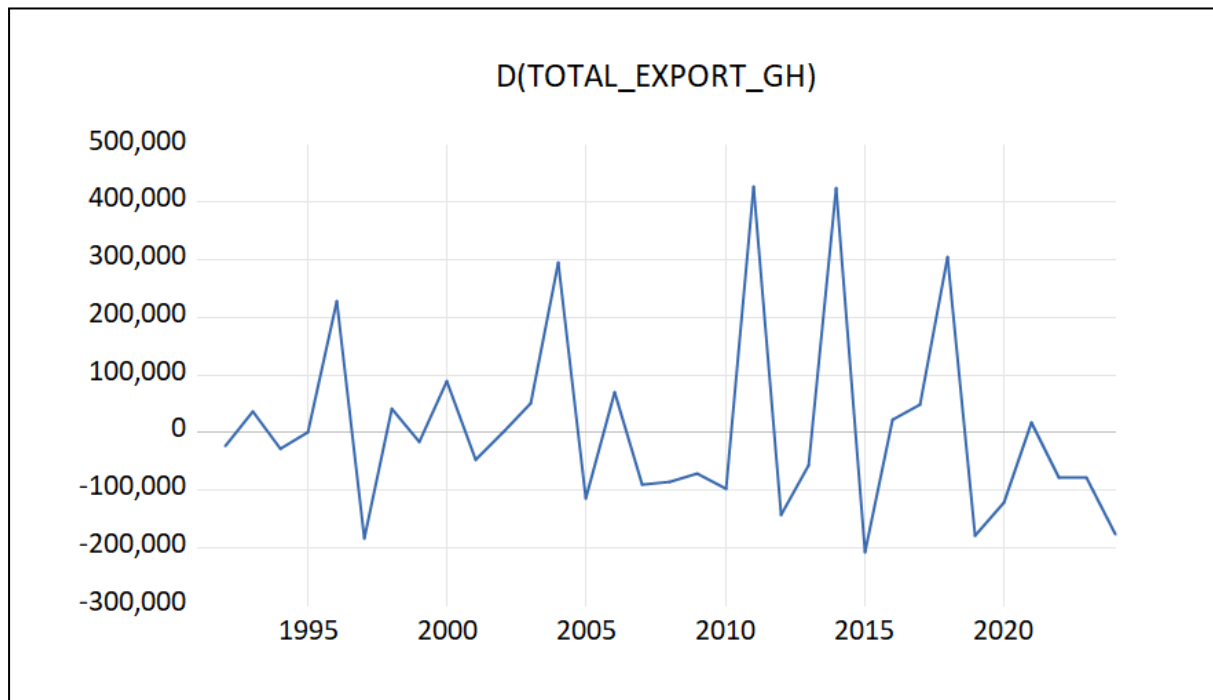




Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: TOTAL EXPORT GH has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-3.206105	0.1007
Test critical values:				
	1% level		-4.262735	
	5% level		-3.552973	
	10% level		-3.209642	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(TOTAL EXPORT GH)				
Method: Least Squares				
Date: 10/07/26 Time: 18:40				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
TOTAL EXPORT GH(-1)	-0.620270	0.193465	-3.206105	0.0032
C	162607.8	64100.54	2.536762	0.0166
@TREND("1991")	11628.27	4979.958	2.335013	0.0264
R-squared	0.264688	Mean dependent var		7792.522
Adjusted R-squared	0.215667	S.D. dependent var		163197.6
S.E. of regression	144532.0	Akaike info criterion		26.68690
Sum squared resid	6.27E+11	Schwarz criterion		26.82294
Log likelihood	-437.3338	Hannan-Quinn criter.		26.73267
F-statistic	5.399505	Durbin-Watson stat		1.895029
Prob(F-statistic)	0.009933			

Como la tendencia determinística resulta significativo analizo el p-valor para verificar la presencia de raíz unitaria al 5% de significatividad, como el p-valor es $0.1007 > 0.05$, no puedo rechazar la hipótesis nula de raíz unitaria y por lo tanto debo diferenciar la serie.



Nuevamente hago los test de Dickey Fuller en busca de raíz unitaria en la serie diferenciada:

Null Hypothesis: D(TOTAL EXPORT GH) has a unit root
 Exogenous: Constant, Linear Trend
 Lag Length: 0 (Automatic - based on SIC, maxlag=8)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-7.703980	0.0000
Test critical values:		
1% level	-4.273277	
5% level	-3.557759	
10% level	-3.212361	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
 Dependent Variable: D(TOTAL EXPORT GH,2)
 Method: Least Squares
 Date: 10/07/26 Time: 18:42
 Sample (adjusted): 1993 2024
 Included observations: 32 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(TOTAL EXPORT GH(-1))	-1.356999	0.176143	-7.703980	0.0000
C	59532.00	60482.02	0.984293	0.3331
@TREND("1991")	-2625.793	3049.421	-0.861079	0.3963

R-squared	0.672435	Mean dependent var	-4767.246
Adjusted R-squared	0.649845	S.D. dependent var	268726.9
S.E. of regression	159016.2	Akaike info criterion	26.88046
Sum squared resid	7.33E+11	Schwarz criterion	27.01787
Log likelihood	-427.0874	Hannan-Quinn criter.	26.92601
F-statistic	29.76608	Durbin-Watson stat	2.140908
Prob(F-statistic)	0.000000		

Como la tendencia determinística no resulta significativa al 5% ($0.39 > 0.05$) pruebo el test con intercepto y sin tendencia:

Null Hypothesis: D(TOTAL EXPORT GH) has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-7.700765	0.0000
Test critical values:	1% level		-3.653730	
	5% level		-2.957110	
	10% level		-2.617434	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(TOTAL EXPORT GH,2)				
Method: Least Squares				
Date: 10/07/26 Time: 18:42				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(TOTAL EXPORT GH(-1))	-1.348405	0.175100	-7.700765	0.0000
C	13464.44	28088.92	0.479351	0.6352
R-squared	0.664060	Mean dependent var		-4767.246
Adjusted R-squared	0.652862	S.D. dependent var		268726.9
S.E. of regression	158329.5	Akaike info criterion		26.84321
Sum squared resid	7.52E+11	Schwarz criterion		26.93481
Log likelihood	-427.4913	Hannan-Quinn criter.		26.87357
F-statistic	59.30178	Durbin-Watson stat		2.101273
Prob(F-statistic)	0.000000			

El intercepto tampoco resulta significativo ($0.63 > 0.05$) por lo tanto hago el test sin tendencia ni intercepto:

Null Hypothesis: D(TOTAL EXPORT GH) has a unit root				
Exogenous: None				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-7.785040	0.0000
Test critical values:	1% level		-2.639210	
	5% level		-1.951687	
	10% level		-1.610579	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(TOTAL EXPORT GH,2)				
Method: Least Squares				
Date: 10/07/26 Time: 18:43				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(TOTAL EXPORT GH(-1))	-1.341331	0.172296	-7.785040	0.0000
R-squared	0.661487	Mean dependent var		-4767.246
Adjusted R-squared	0.661487	S.D. dependent var		268726.9
S.E. of regression	156350.2	Akaike info criterion		26.78834
Sum squared resid	7.58E+11	Schwarz criterion		26.83414
Log likelihood	-427.6134	Hannan-Quinn criter.		26.80352
Durbin-Watson stat	2.095762			

Aqui comparo directamente el t-statistic estimado con el estadístico al 5%, como $-7.88 < -1.95$ rechazo la hipótesis nula de raíz unitaria en la serie diferenciada, por lo que procedo al análisis del correlograma:

Date: 10/07/26 Time: 18:49						
Sample (adjusted): 1992 2024						
Included observations: 33 after adjustments						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	-0.334	-0.334	4.0273	0.045
		2	-0.085	-0.222	4.2982	0.117
		3	0.152	0.051	5.1902	0.158
		4	0.038	0.117	5.2474	0.263
		5	-0.074	0.019	5.4741	0.361
		6	-0.274	-0.356	8.6898	0.192
		7	0.271	0.011	11.954	0.102
		8	-0.002	0.095	11.954	0.153
		9	-0.223	-0.083	14.357	0.110
		10	0.093	-0.057	14.792	0.140
		11	-0.007	-0.138	14.795	0.192
		12	-0.006	-0.072	14.797	0.253
		13	-0.096	-0.005	15.325	0.287
		14	0.087	0.040	15.780	0.327
		15	0.118	0.070	16.678	0.338
		16	-0.036	0.119	16.764	0.401

Aunque el correlograma presenta pocas autocorrelaciones significativas, el test de Ljung-Box para el primer rezago presenta un p-valor de 0,045, por lo que se rechaza al 5% la hipótesis de ausencia de autocorrelación.

Por lo tanto, la serie diferenciada no puede considerarse ruido blanco y se procede a identificar una estructura ARMA.

Dependent Variable: D(TOTAL EXPORT GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 10/07/26 Time: 19:39				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Convergence achieved after 11 iterations				
Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	-0.341331	0.172296	-1.981073	0.0565
R-squared	0.109818	Mean dependent var		8753.678
Adjusted R-squared	0.109818	S.D. dependent var		165714.0
S.E. of regression	156350.2	Akaike info criterion		26.78834
Sum squared resid	7.58E+11	Schwarz criterion		26.83414
Log likelihood	-427.6134	Hannan-Quinn criter.		26.80352
Durbin-Watson stat	2.095762			
Inverted AR Roots	-.34			

El coeficiente ϕ_1 no resultó significativo al 5%, por lo que ahora intento con el componente MA.

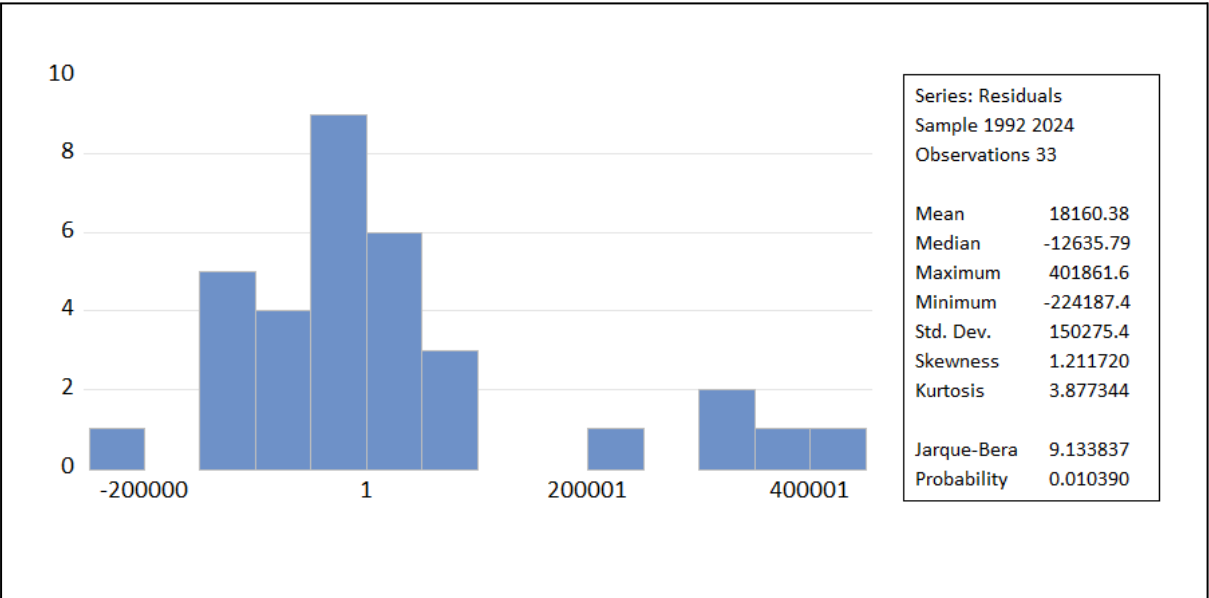
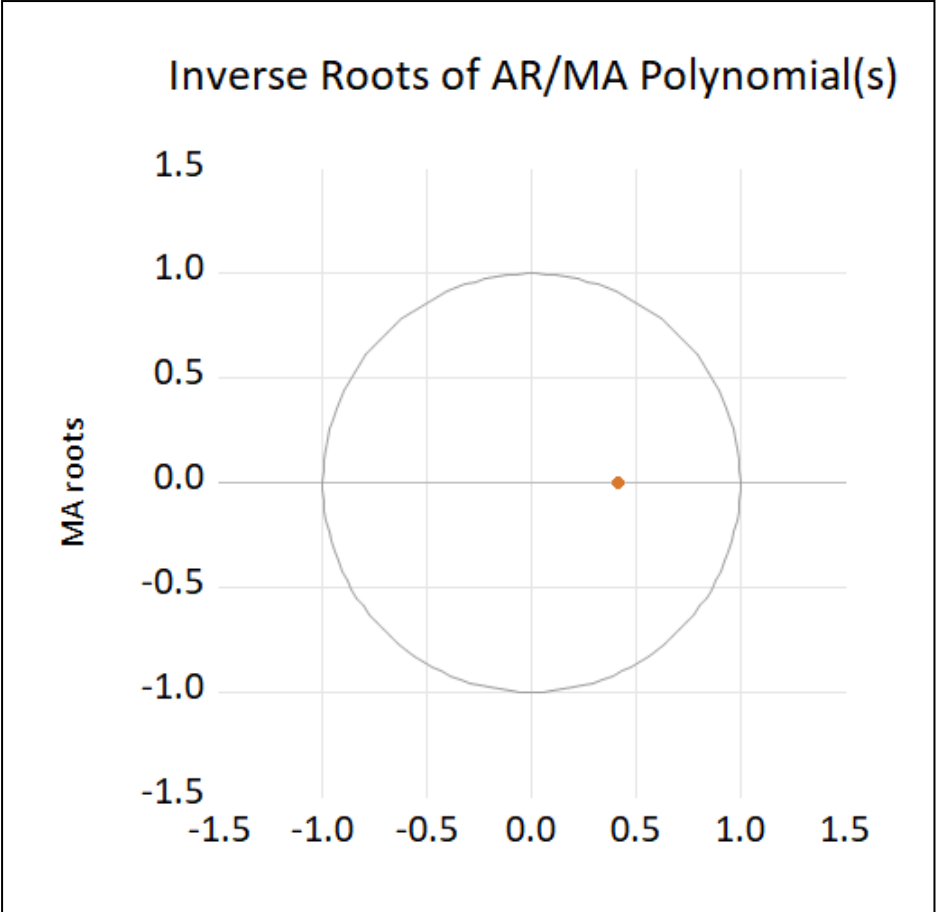
Dependent Variable: D(TOTAL_EXPORT_GH)
 Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
 Date: 10/07/26 Time: 18:51
 Sample (adjusted): 1992 2024
 Included observations: 33 after adjustments
 Failure to improve likelihood (non-zero gradients) after 14 iterations
 Coefficient covariance computed using outer product of gradients
 MA Backcast: 1991

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	-0.413805	0.171422	-2.413954	0.0217
R-squared	0.139323	Mean dependent var	7792.522	
Adjusted R-squared	0.139323	S.D. dependent var	163197.6	
S.E. of regression	151402.8	Akaike info criterion	26.72311	
Sum squared resid	7.34E+11	Schwarz criterion	26.76846	
Log likelihood	-439.9313	Hannan-Quinn criter.	26.73837	
Durbin-Watson stat	1.942983			
Inverted MA Roots	.41			

El coeficiente resulta significativo y la raíz invertida está dentro del círculo unitario, por lo que paso a ver el comportamiento de los residuos:

Date: 10/07/26 Time: 18:57
 Sample (adjusted): 1992 2024
 Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.028	-0.028	0.0280	
		2 -0.063	-0.064	0.1744	0.676
		3 0.124	0.121	0.7684	0.681
		4 0.013	0.015	0.7749	0.855
		5 -0.191	-0.178	2.2735	0.686
		6 -0.298	-0.337	6.0759	0.299
		7 0.198	0.172	7.8254	0.251
		8 -0.009	0.037	7.8295	0.348
		9 -0.228	-0.166	10.340	0.242
		10 -0.003	-0.129	10.340	0.324
		11 0.031	-0.106	10.389	0.407
		12 -0.027	-0.004	10.430	0.492
		13 -0.063	0.072	10.657	0.558
		14 0.171	0.093	12.437	0.492
		15 0.158	0.029	14.048	0.446
		16 -0.028	-0.008	14.102	0.518



El test de Jarque-Bera rechaza la hipótesis nula de normalidad de los residuos al 5%. Esto representa una limitación para los resultados que dependan de una distribución normal de los errores, sin embargo no invalida el resultado obtenido ARIMA(0,1,1) mientras los residuos se comporten como ruido blanco.